

CS6224 WEEK 12

LAB 10

AGENDA

1. Overview
2. Download today lab exercise PKA files:
 - Lab 10a
 - Lab 10b
3. Revision of Wireless Network Risk & Mitigation
4. Briefing of the lab activity
5. Complete the lab exercise
6. Q & A

WIRELESS NETWORK SECURITY RISK & MITIGATION

Security Risk	Description	Solution / Mitigation
Eavesdropping	Attackers intercept unencrypted wireless communication	Use WPA3 or WPA2 encryption- Avoid open/public networks
Rogue Access Points	Unauthorized APs mimic legitimate networks to steal data	Use WIDS/WIPS (Wireless Intrusion Detection/Prevention Systems). Regular network audits
Denial of Service (DoS)	Flooding the network to make services unavailable	Use rate limiting, firewalls, and WIDS
MAC Address Spoofing	Attacker changes their MAC to bypass filters or impersonate devices	Implement MAC filtering with 802.1X authentication
Weak/Default Passwords	Easily guessed passwords allow attackers access to network	Enforce strong, unique passwords- Change default credentials
Outdated Firmware	Old firmware may contain vulnerabilities	Perform regular firmware updates on routers and APs
Evil Twin Attack	Fake AP created with same SSID to lure users	Train users to verify certificates. Use SSID hiding cautiously
Wireless Sniffing Tools	Tools like Wireshark used to capture wireless traffic	Enforce encryption, disable SSID broadcast in some cases

WIRELESS NETWORK DEVICES

Device	Primary Function	Technology Used	Deployment Scenario	Security Features
Wireless Router	Combines routing, switching, and wireless access in one device	Wi-Fi (802.11 a/b/g/n/ac/ax), Ethernet	Home networks, small offices	WPA2/WPA3, NAT, firewall, basic VPN support
Lightweight Access Point (LAP)	Provides wireless access, managed by a WLC	Wi-Fi (802.11 a/b/g/n/ac/ax)	Enterprise networks (schools, hospitals, offices)	No local config; relies on WLC for policy enforcement
Wireless LAN Controller (WLC)	Centralizes control and management of multiple lightweight APs	Control and Provisioning of Wireless Access Points (CAPWAP)	Large-scale enterprise networks	Centralized security policies, RADIUS/TACACS+, WPA3
Wireless Access Point (Autonomous)	Connects wireless devices directly to the wired network	Wi-Fi	Small businesses, legacy systems	WPA2/WPA3, local ACLs, SSID control
Wireless Range Extender	Boosts and repeats wireless signal	Wi-Fi	Homes with weak signals	WPA2/WPA3, basic pass-through encryption
Mobile Hotspot	Shares cellular internet over Wi-Fi	4G/5G LTE + Wi-Fi	Travel, field deployments	WPA2, basic firewall, device limits
Bluetooth Adapter	Enables short-range device communication	Bluetooth 4.0/5.0	Personal area networks, peripherals	Pairing authentication, frequency hopping
Wireless Sensor Node (WSN)	Collects data and transmits wirelessly in IoT applications	Zigbee, BLE, Wi-Fi	Smart homes, wearables, industrial IoT	Lightweight encryption, key exchange
Wearable IoT Device	Collects and transmits physiological or contextual data	BLE, Wi-Fi, Zigbee	Health monitoring, fitness, smartwatches	Lightweight cryptography, mutual authentication

**THANK
YOU**